SPARKY'S SCIENCE EXPERIMENTS

Lemon Battery: Electricity from Fruit

Physical Science (electricity) · Ages 8+ · 45 minutes

Make a battery from lemons, copper and zinc-coated nails that lights a small LED.

Safety first

Sparky always says: Never connect your lemon battery to a wall plug socket - that is very dangerous!

- An adult must help - nails are sharp.
- Don't eat the lemons afterwards - metal gets into the juice.

You need

- 3 or 4 juicy lemons (potatoes work too)
- 3 or 4 pieces of copper: bare copper wire, or 10c, 20c or 50c coins (copper- or bronze-coated)
- 3 or 4 galvanised (zinc-coated) nails
- 5 crocodile-clip wires
- 1 small red LED

Steps

1. Roll each lemon firmly on the table to make it juicy inside.
2. Push one copper piece and one nail into each lemon, 2 cm apart. They must not touch.
3. Make a chain: a wire from the copper of lemon 1 to the nail of lemon 2, and so on.
4. Clip the copper at one end of the chain to the LED's long leg (+), and the nail at the other end to the short leg (-).
5. Switch off the lights and look closely: the LED glows faintly!

If it doesn't work, try these in order

1. Turn the LED round - this is the most common mistake.
2. Make sure the coin and nail don't touch inside each lemon.
3. Check every clip is tight, then add another lemon. Look in a dark room - the glow is faint.

Why it works

The electricity comes from two different metals. The zinc slowly dissolves in the lemon juice and gives away tiny particles called electrons, which flow through the wire and LED to the copper: an electric current. The zinc nail is the negative side (anode) and the copper is the positive side (cathode). The sour juice is the electrolyte. One lemon gives about 1 volt and an LED needs about 2, so we join lemons in series.

Did you know? A torch battery works the same way - two different materials and an electrolyte - just much stronger.

Ask Sparky: Would an orange or a potato work? What happens with only one lemon?